PROPER MODIFICATIONS ALGEBRAIZE TO FORMAL MODIFICATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-modification-gives-formal-modification; the proof environment immediately following it is its own proof

Editorial note. The label lemma-modification-gives-formal-modification names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the definition of a formal modification f_{/T'} : X'_{/T'} to X_{/T} via the four conditions: locally Noetherian source and target, properness, rig-etale-ness, and rig-surjectivity including on diagonals (section-completion-functor setup)
- the four lemmas of the same chapter: proper-gives-proper, etale-gives-rig-etale, completion-proper-surjective-rig-surjective, and separated-mono-open-diagonal-rig-surjective
- the Formal Spaces lemma that formal completions of locally Noetherian schemes are locally Noetherian formal algebraic spaces (formal-spaces-lemma-formal-completion-types)

Source reference. Original lemma 0AQ1: sources/stacks-chow/tags.txt, source lines 13494–13494. Prerequisite pointer invoked at proof line(s) 6355; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AQW: sources/stacks-restricted/pinned-original-source.tex, source lines 6201–6201. Prerequisite pointer invoked at proof line(s) 6361; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AR2: sources/stacks-restricted/pinned-original-source.tex, source lines 6186–6186. Prerequisite pointer invoked at proof line(s) 6360; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GDD: sources/stacks-restricted/pinned-original-source.tex, source lines 6053–6053. Prerequisite pointer invoked at proof line(s)

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6359; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GDI: sources/stacks-restricted/pinned-original-source.tex, source lines 6241–6241. Prerequisite pointer invoked at proof line(s) 6362; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0GDB: sources/stacks-restricted/pinned-original-source.tex, source lines 5953–5953. Prerequisite pointer invoked at proof line(s) 6349; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S, $f : X' \to X$, $T \subset |X|$, $U \subset X$, $T' \subset |X'|$, and $U' \subset X'$ be as in Section 0GDB. If X is locally Noetherian, f is proper, and $U' \to U$ is an isomorphism, then $f_{/T} : X'_{/T'} \to X_{/T}$ is a formal modification.*

Proof. By Formal Spaces, Lemmas 0AQ1 the source and target of the arrow are locally Noetherian formal algebraic spaces. The other conditions follow from Lemmas 0GDD, 0AR2, 0AQW, and 0GDI. □